

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11 – SUB-STRAND 1.1 TAXONOMY I: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the whole unit. Our driving question was: How can we organise the many different living organisms around us so that everyone can identify and study them in the same way? Think back to Lesson 1, when your group sorted a goat, a chicken, a maize plant, a mushroom, algae, bacteria, a bean plant and a mosquito, and found that other groups sorted them differently and used different names in Kikuyu, Luo, Kiswahili and English. Answer each section in the space under the prompt, using full sentences. Use evidence from the activities you did (sorting, case sets, rank cards, name comparison cards, kingdom profile cards). Use the Claim-Evidence-Reasoning (CER) structure where the prompt asks for it. Write scientific names correctly. Your group poster or classification guide of local organisms and your own written explanation will be marked on: correct use of characteristics, evidence and justification, accuracy of classification and naming, communication and collaboration, and local application and conservation.

Part 1: Why our first groupings were not enough

Look back at the Lesson 1 sorting activity with the eight organisms. (a) Describe two different ways groups sorted the organisms and what you noticed when groups compared. (b) Why was every group correct by its own rules, yet the groupings could not be used by everyone in the same way? (c) Explain why local names alone are not enough for scientists, using what you found about names for the same organism in Lesson 1.

Write your answer here:

Part 2: The hierarchy of taxonomic units

(a) List the seven taxonomic ranks in order from the broadest to the most specific. (b) Choose one Kenyan organism such as the goat or the maize plant (*Zea mays*) and follow it from Kingdom down to Species. Explain what happens to the number of organisms in a group and how similar they are as you move down. (c) Write a CER statement: Claim, Evidence and Reasoning for why a ranked hierarchy helps people in Nairobi, Kisumu and Mombasa identify an organism in the same way.

Write your answer here:

Part 3: One name for every organism

(a) Explain why common names cause problems when people share information, using maize, beans, goat or mosquito as an example. **(b)** State the rules for writing a scientific name using binomial nomenclature. **(c)** Here are three names written wrongly: 'zea Mays', 'capra hircus' and 'Anopheles Gambiae'. Rewrite each correctly and say which rule was broken in each name.

Write your answer here:

Part 4: Placing organisms into the five kingdoms

Choose any four of these organisms from the final challenge: Spirogyra, Escherichia coli, Rhizopus on stale bread, Agaricus (mushroom), Zea mays, tilapia (Oreochromis niloticus), Anopheles mosquito, Amoeba, Acacia. For each one, name its kingdom out of the five kingdoms, and give the observable evidence (such as cell type, number of cells, cell wall, chlorophyll and mode of nutrition) that supports your decision. Then explain why these characteristics are better for classifying than colour, habitat or usefulness.

Write your answer here:

Part 5: Our final explanation and why taxonomy matters

Write your Final Explanation answering the driving question: How can living organisms be systematically organised, named and identified? Use a CER structure. Include: how organisms are grouped, how they are named, how you can identify the kingdom of an organism, and why this matters for people in Kenya (for example disease vectors, crop pests, useful organisms, and conserving biodiversity). Finally, say how your thinking has changed from your Lesson 1 model.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Correct use of characteristics (30%)	Consistently uses observable characteristics (cell type, number of cells, cell wall, chlorophyll, mode of nutrition) correctly to place organisms in kingdoms and explains why they are better than colour, habitat or usefulness.	Uses most characteristics correctly with minor errors or incomplete explanation of why they are better than personal criteria.	Uses few or incorrect characteristics, or relies on colour, habitat or usefulness.
Evidence and justification (25%)	Claim-Evidence-Reasoning is complete, with specific evidence from the lessons (sorting, case sets, rank cards, name cards, profile cards) linked clearly to the claim.	CER is present with some evidence from the lessons, but the reasoning is partly missing or general.	Claim is given with little or no evidence or reasoning.
Accuracy of classification and naming (20%)	Ranks are in the correct order, kingdoms are correct, and scientific names follow	Mostly accurate with one or two errors in ranks, kingdoms or name writing.	Several errors in ranks, kingdoms or the writing of scientific names.

	all rules (capital genus, lower-case species, italics or underlined, two parts).		
Communication and collaboration (15%)	Writing and group poster are clear, logical and well organised, using correct vocabulary, with clear evidence of contributing to the group.	Writing is mostly clear with some correct vocabulary; contribution to group work is shown.	Writing is unclear or disorganised, with little correct vocabulary or little evidence of group contribution.
Local application and conservation connection (10%)	Uses Kenyan organisms and places and clearly explains how taxonomy helps with disease vectors, crop pests, useful organisms and conserving biodiversity.	Gives a Kenyan example and a general link to the importance of taxonomy or conservation.	Little or no link to Kenyan examples or the importance of taxonomy.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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